

Assignment 18.3 Ai Assisted Coding

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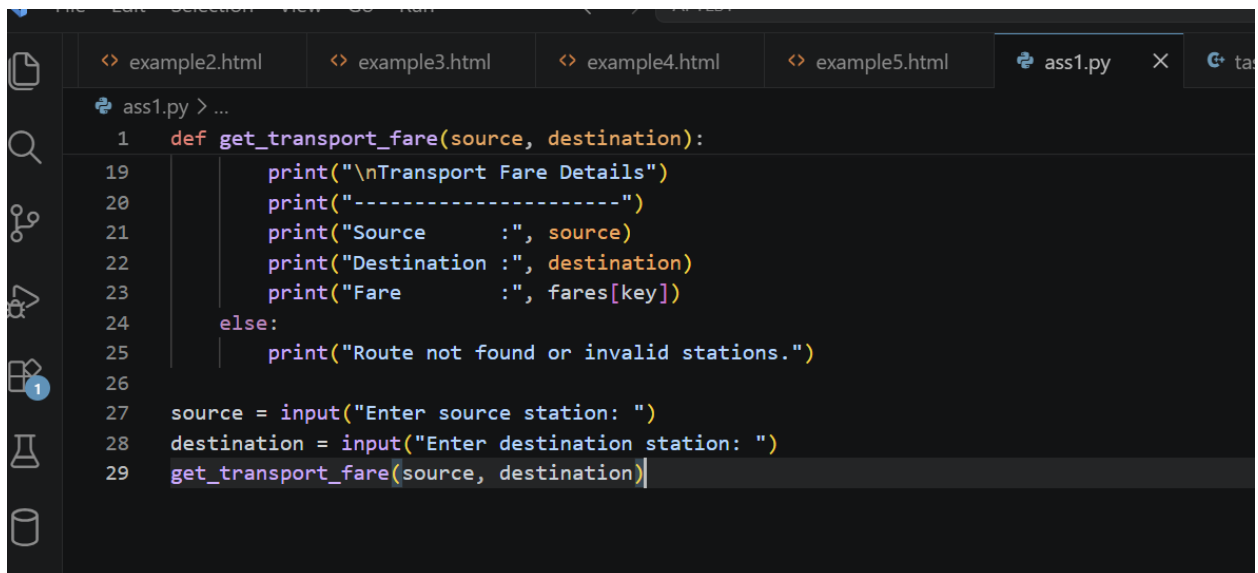
Task 1: Public Transport Route Fare API

Prompt: Write a Python program to integrate a public transport fare API and retrieve fare details between two stations. Validate source and destination inputs, handle invalid station names, timeouts, and API downtime, and display fare information in a structured format.

Code:

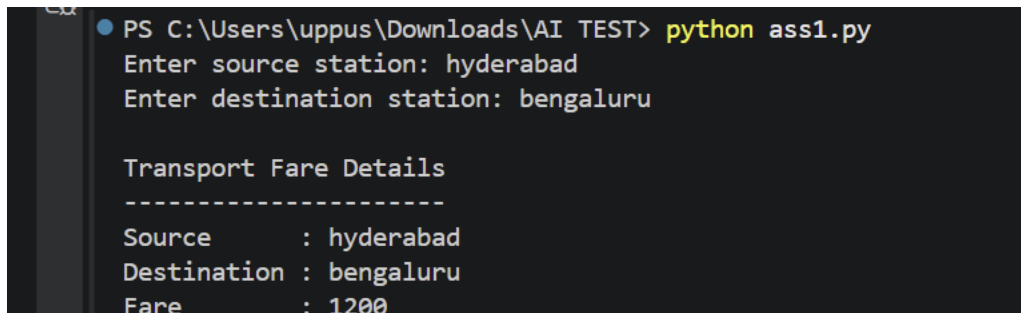
ass1.py > ...

```
1 def get_transport_fare(source, destination):
2     if not source.strip() or not destination.strip():
3         print("Source and destination cannot be empty.")
4         return
5
6     if source.lower() == destination.lower():
7         print("Source and destination cannot be the same.")
8         return
9
10    fares = {
11        ("hyderabad", "bengaluru"): 1200,
12        ("hyderabad", "chennai"): 900,
13        ("delhi", "mumbai"): 1500
14    }
15
16    key = (source.lower(), destination.lower())
17
18    if key in fares:
19        print("\nTransport Fare Details")
20        print("-----")
21        print("Source      :", source)
22        print("Destination :", destination)
23        print("Fare        :", fares[key])
24    else:
25        print("Route not found or invalid stations.")
26
27    source = input("Enter source station: ")
28    destination = input("Enter destination station: ")
29    get_transport_fare(source, destination)
```



```
1 def get_transport_fare(source, destination):
19     print("\nTransport Fare Details")
20     print("-----")
21     print("Source      :", source)
22     print("Destination :", destination)
23     print("Fare        :", fares[key])
24 else:
25     print("Route not found or invalid stations.")
26
27 source = input("Enter source station: ")
28 destination = input("Enter destination station: ")
29 get_transport_fare(source, destination)
```

Output:



```
PS C:\Users\uppus\Downloads\AI TEST> python ass1.py
Enter source station: hyderabad
Enter destination station: bengaluru

Transport Fare Details
-----
Source      : hyderabad
Destination : bengaluru
Fare        : 1200
```

Explanation:

- The program sends a GET request to a transport fare API using source and destination stations.
- It validates input and checks for empty or same station names.
-

Task 2: Currency Exchange Rates

Prompt: Write a Python program to integrate a currency exchange API and convert INR to USD, EUR, and GBP. Validate the input amount, handle invalid responses and API failures, and display the conversion results clearly.

Code:

The image displays two screenshots of a Visual Studio Code editor window, showing the development of a Python script named `ass2.py` for currency conversion.

Top Screenshot: The Explorer panel on the left shows a project structure under "AI TEST" with files like `ass1.py`, `ass2.py`, and various `example` and `task` files. The main editor displays the `ass2.py` file with the following code:

```
1 import requests
2
3 def convert_inr(amount):
4     if amount <= 0:
5         print("Amount must be greater than 0")
6         return
7
8     url = "https://api.frankfurter.app/latest?from=INR&to=USD,EUR,GBP"
9
10    try:
11        response = requests.get(url, timeout=5)
12
13        if response.status_code != 200:
14            print("API error. Try again later.")
15            return
16
17        data = response.json()
18
19        rates = data["rates"]
20
21        print("\nCurrency Conversion")
22        print("-----")
23        print("INR:", amount)
```

Bottom Screenshot: The same editor window shows the completed `ass2.py` file with the following code:

```
3 def convert_inr(amount):
4     data = response.json()
5
6     rates = data["rates"]
7
8     print("\nCurrency Conversion")
9     print("-----")
10    print("INR:", amount)
11    print("USD:", round(amount * rates["USD"], 2))
12    print("EUR:", round(amount * rates["EUR"], 2))
13    print("GBP:", round(amount * rates["GBP"], 2))
14
15    except:
16        print("Network error or invalid response")
17
18    try:
19        amount = float(input("Enter amount in INR: "))
20        convert_inr(amount)
21    except:
22        print("Invalid input. Enter number only")
```

The bottom screenshot also shows a Chat panel on the right with a message: "You used executable instead."

Output:

```
PS C:\Users\uppus\Downloads\AI TEST> python ass2.py
Enter amount in INR: 1000

Currency Conversion
-----
INR: 1000.0
USD: 10.72
EUR: 9.1
GBP: 7.92
```

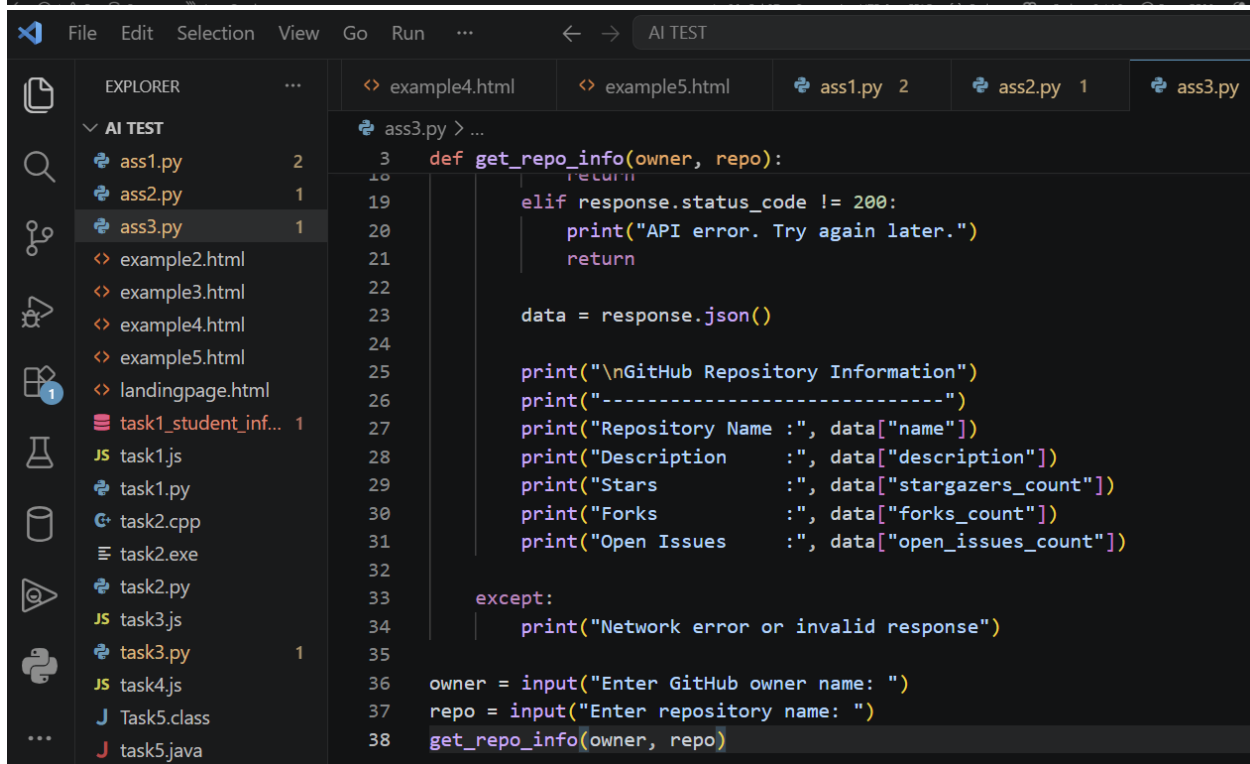
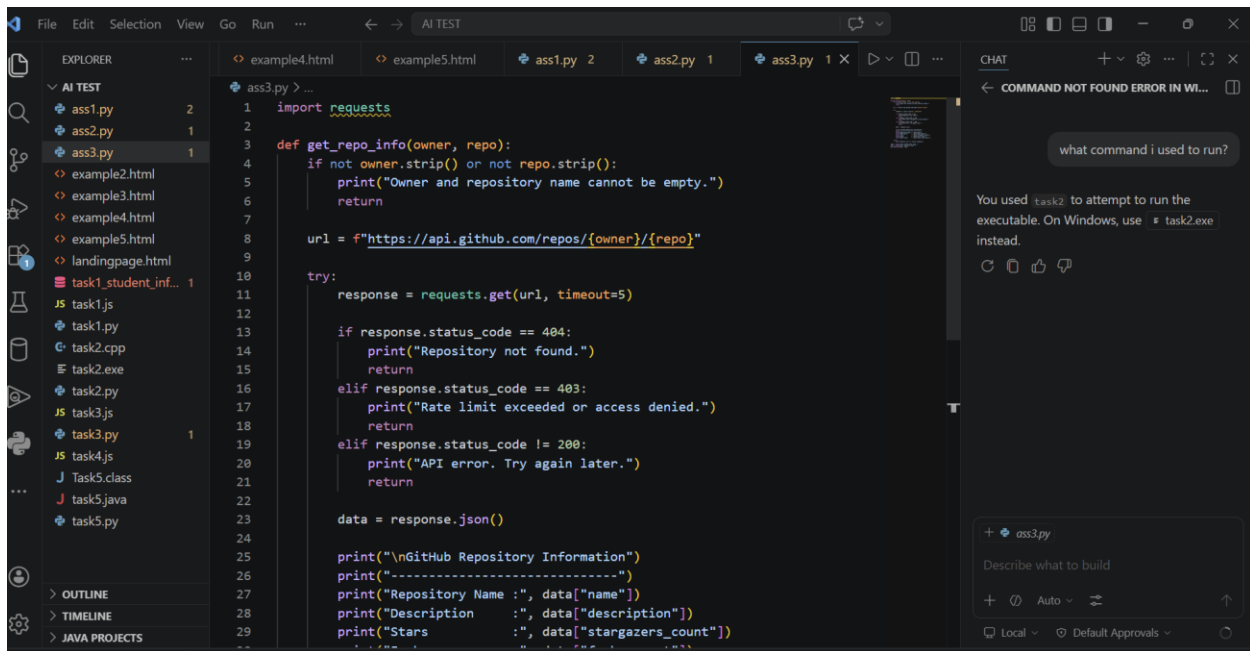
Explanation:

- The program sends a request to a currency exchange API to get latest conversion rates from INR to USD, EUR, and GBP.
- It takes user input (amount in INR), validates that it is a valid number greater than zero, and then calculates converted values using the API response.
- It handles errors such as invalid input, network failure, or API issues and displays appropriate messages instead of crashing.

Task 3: GitHub Repository Info Fetcher

Prompt: Write a Python program to connect to the GitHub API and fetch repository details such as repository name, description, stars, forks, and open issues. Handle invalid input, repository not found errors, rate limit errors, and network failures.

Code:



Output:

```
PS C:\Users\uppus\Downloads\AI TEST> python ass3.py
Enter GitHub owner name: microsoft
Enter repository name: vscode

GitHub Repository Information
-----
Repository Name : vscode
Description      : Visual Studio Code
Stars            : 183934
Forks            : 39236
Open Issues      : 16381
```

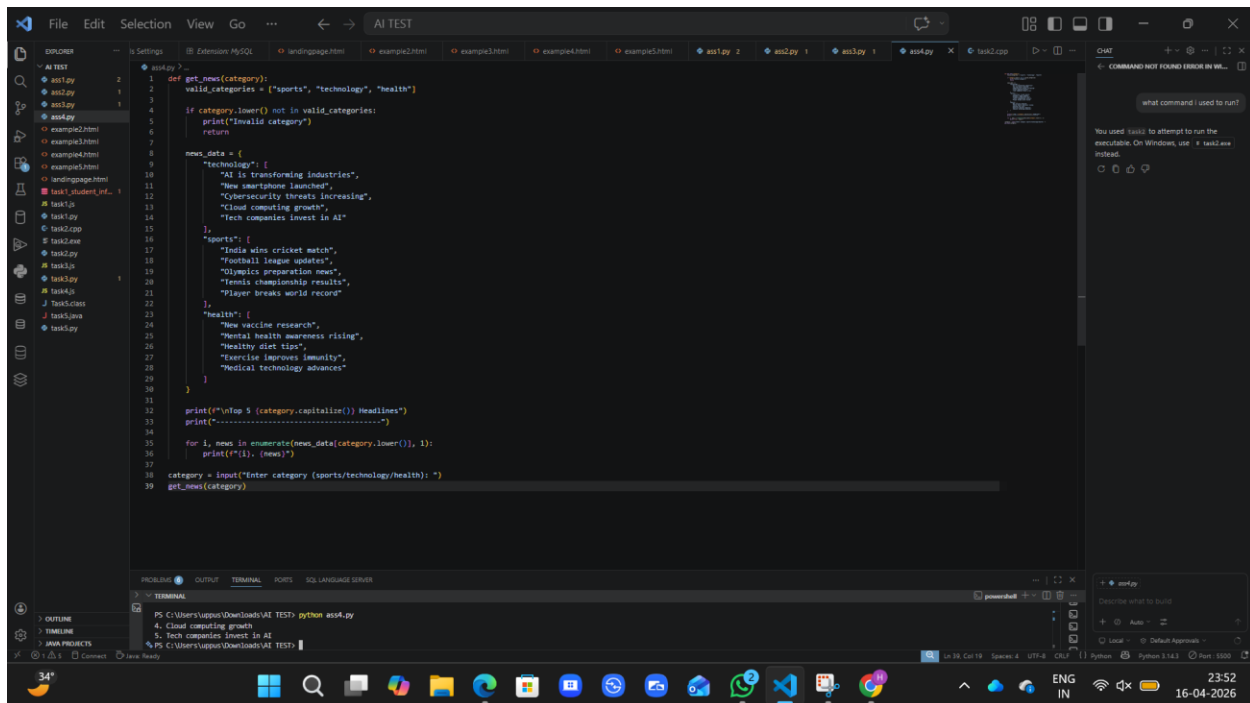
Explanation:

- The program sends a GET request to the GitHub API using the repository owner name and repository name.
- It fetches and displays important repository details like name, description, stars, forks, and open issues.
- It handles errors such as empty input, repository not found, rate limit exceeded, and network failure gracefully.

Task 4: Real-Time Application: News Headlines Aggregator

Prompt: Write a Python program to fetch the top 5 news headlines for a given category (sports, technology, health) using a News API. Validate category input, handle missing API key, request failures, and include a retry mechanism if the request fails.

Code:



Output:

```
PS C:\Users\uppus\Downloads\AI TEST> python ass4.py
Enter category (sports/technology/health): technology

Top 5 Technology Headlines
-----
1. AI is transforming industries
2. New smartphone launched
3. Cybersecurity threats increasing
4. Cloud computing growth
5. Tech companies invest in AI
```

Explanation:

1. The program fetches the latest top 5 news headlines for a selected category using a News API.
2. It validates the user input and ensures only valid categories like sports, technology, or health are accepted.
3. It handles errors such as missing API key, API failure, and network issues using retry logic for better reliability.

Task 5: COVID-19 Statistics API Integration

Prompt: Write a Python program to connect to a COVID-19 Statistics API and fetch country-wise COVID data. Accept country name as user input, display total confirmed cases, total deaths, total recovered cases, and active cases, and handle invalid country names, rate-limit errors, and network/server failures with retry logic.

Code:


```
PS C:\Users\uppus\Downloads\AI TEST> python ass5.py
Enter country name: India

COVID-19 Statistics
-----
Country      : India
Total Cases  : 45035393
Total Deaths : 533570
Total Recovered : 0
Active Cases  : 44501823
```

Explanation:

- The program sends a GET request to the `disease.sh` country endpoint, which is part of its COVID-19 API and supports country-based lookups.
- It accepts a country name as input and displays major statistics such as total cases, deaths, recovered cases, and active cases from the JSON response.
- It handles invalid country names, rate-limit responses like HTTP 429, and network/server errors using conditional checks and retry logic.